JNTUH USED 04-05-17 PM Set 2

R13

Code No: 118EQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, May - 2017
SEMANTIC WEB AND SOCIAL NETWORKS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) Define web. [2]
b) Give the limitations of today's web. [3]
c) What is XQUEY? [2]
d) What is the use of reification mechanism in RDF? [3]
e) Give modus tollens rule with an example. [2]
f) How is knowledge base different from database? [3]
g) What is the role of SOAP in web services? [2]
h) Give OWL-S process ontology. [3]
i) What is Jaccard coefficient? [2]
j) What influence does social science has on social network analysis? [3]

PART - B

(50 Marks)

2. Make a comparison of industrial revolution and information revolution. Explain the significance of web in information revolution. [10]

OR

3. What are the objectives of semantic web? Discuss briefly the role software agents and inference engines in semantic web. [10]

4.a) Can XML replace HTML? Justify your answer.
b) With illustrations explain inheritance in RDF. [5+5]

OR

5.a) What are the requirements of ontology language?
b) Discuss Learning Source Description tool for ontology mapping. [5+5]

6.a) Quote examples for monotonic rules and non monotonic rules. How are they used in semantic web?
b) Discuss descriptive logic for semantic web. [5+5]

OR

7. What is a web service? Is it different from web application? What is the need of semantic technology for web services? Discuss semantic web services. [10]

8.a) Give the functional description parameters of Service Profile.
b) Why is process ontology control is still an area under significant development? [5+5]

OR

9.a) What is a sociogram? Why is it required?
b) Why electronic discussion networks become important source for social network analysis? [5+5]

10. How do blogs and online communities provide data and help in social network analysis? Can this data help predictions? Justify your answer. [10]

OR

11. Explain the following features of Elmo scutter:
a) Blacklisting
b) Whitelisting
c) Filtering
d) Persistence
e) Logging. [10]

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